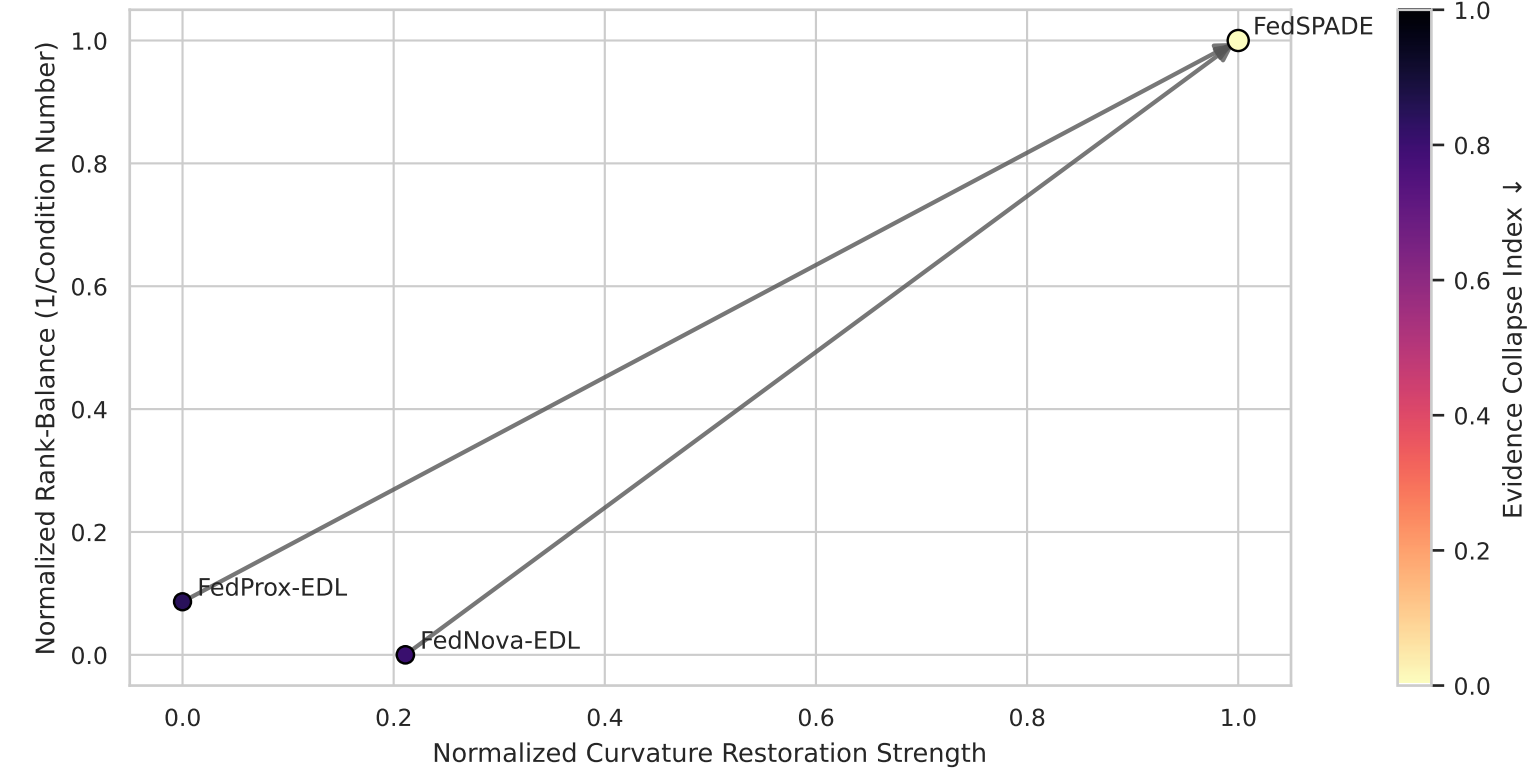
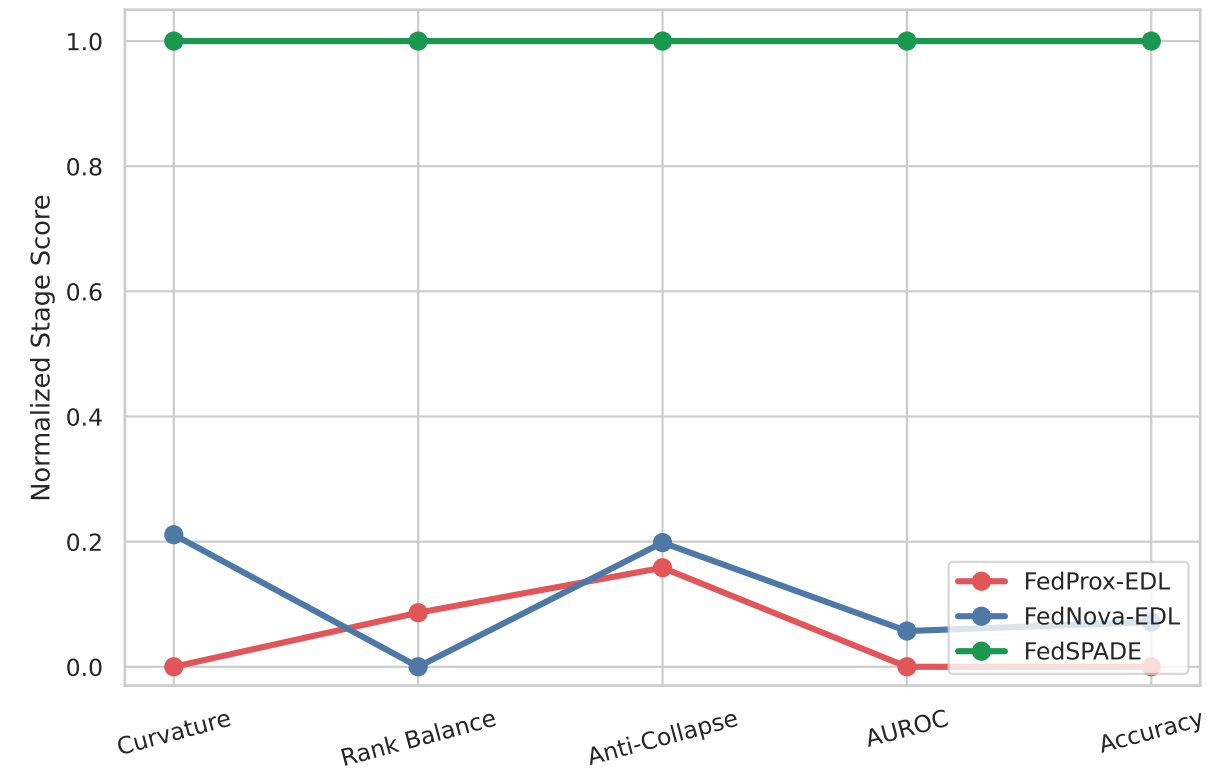


FedSPADE Ablation: Curvature → Rank → Evidence-Collapse Repair → Reliability/Accuracy

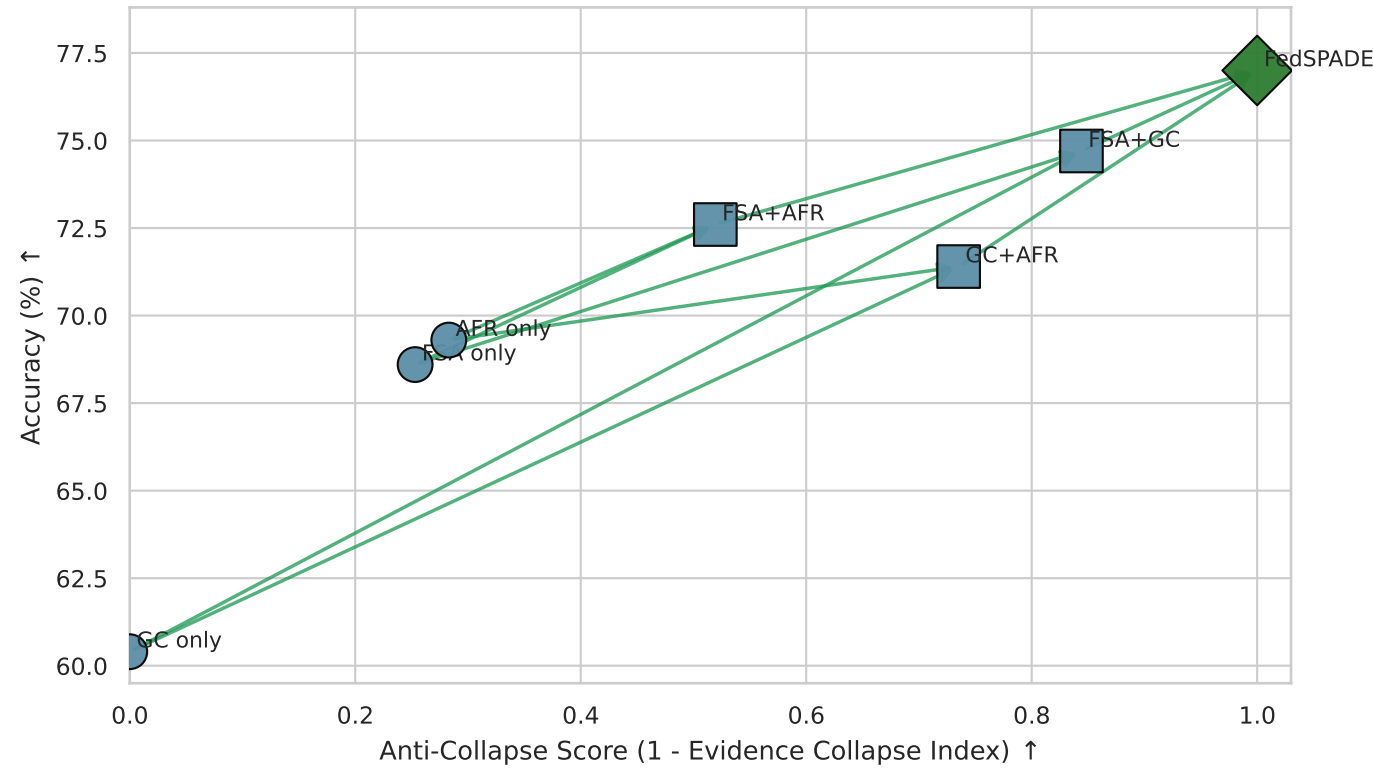
(a) Geometry State Map: Curvature → Rank



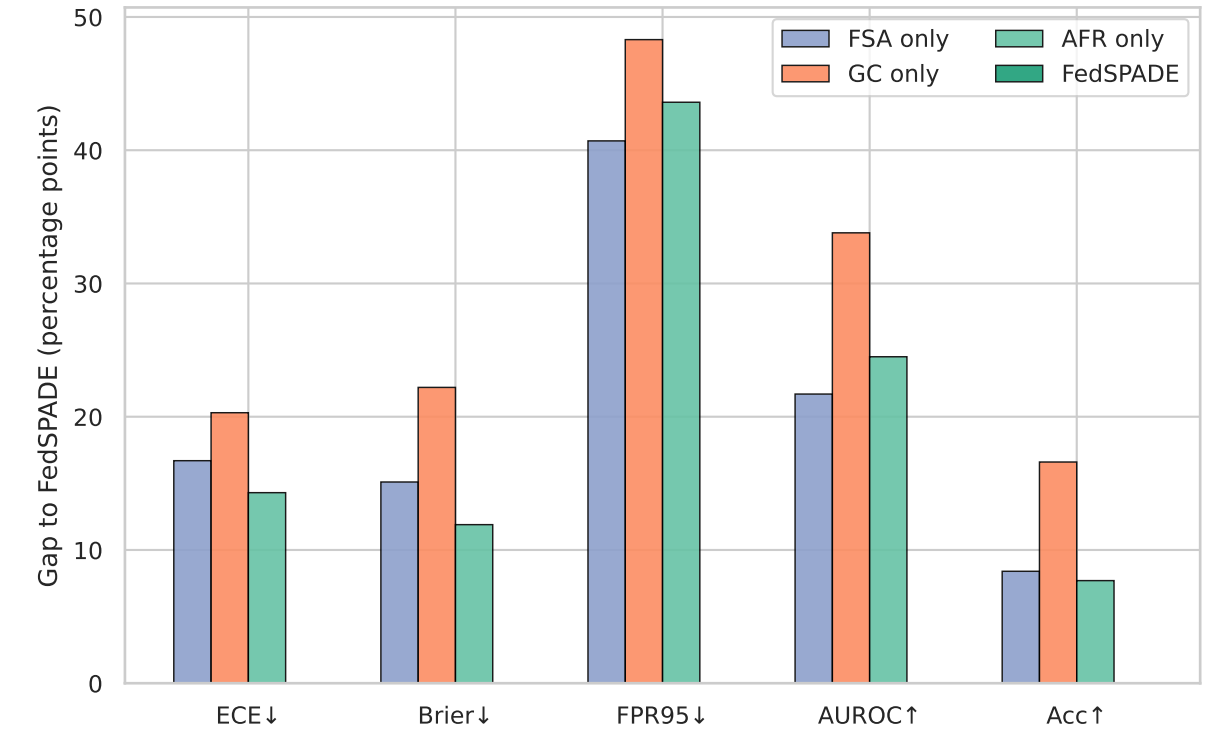
(b) Theory Chain Profile



(c) Ablation Causal Lattice: Collapse Repair → Accuracy



(d) Collapse-Sensitive Metric Gaps



Inputs use reported real values: ablation table, stabilizing-EDL table, and curvature summary (AUC $\lambda_{\min}+$, condition number).